

Contributions of Subregions of the Prefrontal Cortex to Working Memory: Evidence from Brain Lesions in Humans

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Abstract

■ We investigated working memory in patients with focal brain damage involving subregions of the prefrontal cortex (PFC). Lesions in the dorsal portion of lateral PFC or the ventromedial portion of orbital PFC did not impair performance in tasks that required maintenance and monitoring of object or spatial information. Larger lesions involving both

ventral and dorsal parts of the lateral PFC impaired maintenance and monitoring of spatial and object information, with more severe deficits observed in the spatial tasks. The results support a distributed localization of function in lateral PFC during working memory. ■

INTRODUCTION

Working memory refers to the ability to hold, manipulate, and monitor information on-line. Extensive research supports a key role of prefrontal cortex (PFC)—the brain region anterior to the motor association cortex—in working memory. This research includes lesion and neurophysiological studies in nonhuman primates (e.g., Funahashi, Bruce, & Goldman-Rakic, 1989, 1993; Goldman-Rakic, 1987; Passingham, 1985; Mishkin & Pribram, 1956; Jacobsen, 1936), studies of humans with lesions in the PFC (e.g., Barcelo, Suwazono, & Knight, 2000; Nielsen-Bohlman & Knight, 1999; Stone, Baron-Cohen, & Knight, 1998; Petrides, 1997; Owen, Morris, Sahakian, Polkey, & Robbins, 1996; Chao & Knight, 1995, 1998; Owen, Sahakian, Semple, Polkey, & Robbins, 1995), and functional imaging studies showing prefrontal activation during working memory tasks in healthy humans (e.g., Cohen et al., 1997; Courtney, Ungerleider, Keil, & Haxby, 1996; Owen, Milner, Petrides, & Evans, 1996; Smith et al., 1995; McCarthy et al., 1994; Jonides et al., 1993). However, there is an ongoing debate about the functional specialization of subdivisions of the PFC for working memory. Anatomically, PFC can be subdivided into at least three regions (Crespo-Facorro et al., 1999): the ventromedial region involving the straight gyrus, orbitofrontal gyri, and medial frontal cortex (Brodmann's

areas [BA] 10, 11, 12); the dorsolateral PFC including the middle frontal gyrus and part of the superior frontal gyrus (BA 9, 46, 9/46); and a ventrolateral region composed of the inferior frontal gyrus (BA 44, 45, 47). While most authors agree on the anatomical distinction, they differ as to the precise functions ascribed to the named areas.

Two principal theories about the contributions of the different subareas of PFC to working memory have emerged. The “type of information” theory (O’Scaidhe, Wilson, & Goldman-Rakic, 1997; Goldman-Rakic, 1995, 1996a, 1996b) favors a subspecialization driven by stimulus category. In this formulation, the dorsolateral PFC is concerned with spatial memory content whereas more ventral areas are involved in memory for object information. This model is an extension of the known subdivision of posterior brain areas into two information-dependent processing pathways. Both pathways originate in the visual cortex, with object-based information entering a ventral processing stream projecting to inferior temporal cortex and with spatial information being processed along a dorsal pathway projecting to the parietal cortex. These anatomical and functional dichotomies in the posterior cortices are supported by lesion studies with nonhuman primates (Ungerleider & Mishkin, 1982) and neurological patients (Newcombe, Ratcliff, & Damasio, 1987; Levine, Warach, & Farah, 1985). These studies revealed deficits in memory for objects after temporal lobe lesions and impairments in spatial information processing after parietal brain lesions. More recently, this dissociation has been shown in electrophysiological (Mecklinger & Müller, 1996) and functional imaging

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studies (Baker, Frith, Frackowiak, & Dolan, 1996; Kohler, Kapur, Moscovitch, Winocur, & Houle, 1995; Moscovitch, Kapur, Köhler, & Houle, 1995; Haxby et al., 1991).

Neural connections between inferior temporal cortex and ventral PFC and between superior parietal cortex and dorsolateral PFC have also been identified (Barbas & Pandya, 1989; Barbas, 1988; Goldman-Rakic, 1987; Kawamura & Naito, 1984). This anatomical organization suggests that the distinct processing of object and spatial information in different posterior brain regions remains modular within PFC. The type of information theory is also supported by single-unit recording studies in nonhuman primates. Wilson, O'Scalaidhe, and Goldman-Rakic (1993) recorded neural activity from the principal sulcus and the inferior convexity of lateral PFC in nonhuman primates, the proposed homologues of the human dorso- and ventrolateral prefrontal cortices, respectively. The animals performed two versions of a delayed matching-to-sample task: in the nonspatial object task, the monkeys had to keep object shape in mind; in the spatial task, they had to remember a position over a delay. The object memory task increased single-unit activity during the delay period in inferior convexity neurons. Conversely, memory for spatial information correlated with increased discharge in neurons within the principal sulcus. Thus, Goldman-Rakic and her colleagues argued that "the PFC is functionally compartmentalized with respect to the nature of its inputs" (O'Scalaidhe et al., 1997, p. 1135). This model has been bolstered by the results of some imaging studies that observed activation of subregions of the PFC depending on the type of information to be held in working memory (Courtney et al., 1996; McCarthy et al., 1996). However, the activated areas often showed some overlap, for example, memory for faces activated right prefrontal areas in the dorsolateral, ventrolateral, and ventromedial regions in the study of Courtney et al. (1996).

An alternative theory derived from monkey single-unit (Miller, 1999; Rainer, Asaad, & Miller, 1998b; Rao, Rainer, & Miller, 1997) and functional imaging studies (D'Esposito, Aguirre, et al., 1998; D'Esposito, Postle, Ballard, & Lease, 1999; Owen, 1997; Owen et al., 1999; Petrides, 1994, 1995, 1996) and has been proposed based on the "type of processing" necessary for task completion. In this model, the degree to which memory content has to be monitored or manipulated, rather than the nature of the memory content, determines which prefrontal areas are activated. Specifically, ventrolateral parts of PFC would subserve active maintenance of information within memory, whereas dorsolateral PFC would be recruited when additional "monitoring" and "manipulation" within working memory are required. In this theory, activation of different areas would be independent of type of in-

formation. In accordance with this model, activation of the ventral PFC was observed in tasks demanding the active retrieval of object as well as spatial information from memory (Baker et al., 1996; Courtney et al., 1996; Goldberg, Berman, Randolph, Gold, & Weinberger, 1996; Smith et al., 1995; Jonides et al., 1993) and the dorsolateral PFC was activated in nonspatial working memory tasks provided that the tasks made further demands on information processing, such as self-ordering (Petrides, Alivisatos, Evans, & Meyer, 1993; Petrides, Alivisatos, Meyer, & Meyer, 1993), simultaneous accomplishment of two tasks (D'Esposito et al., 1995), or *n*-back tasks (D'Esposito, Aguirre, et al., 1998; Smith, Jonides, & Koeppe, 1996; Cohen et al., 1994). Single-unit data provide some additional support for the type of processing model since individual neurons responsive to both "what" and "where" information have been reported in monkey PFC (Rainer, Asaad, & Miller, 1998a; Rao et al., 1997). This finding suggests that monkey PFC is involved in the integration of sensory information across categories. These data do not discount modular organization of PFC, but do suggest that there is processing in PFC that is independent of stimulus category.

The type of information and the type of processing models make different predictions about the deficits to be expected after damage in ventral or dorsal regions of the PFC. We tested these predictions by studying patients with lesions located in the different regions of the PFC (see Figure 1). The lesions either involved the ventromedial PFC (VM group: BA 10, 11, 12), the dorsolateral PFC alone (DL group: BA 9, 46, 9/46), or dorsal and ventral regions of the lateral PFC (DVL group: BA 9, 46, 9/46, 45, 47). Patients and healthy controls performed four tasks (see Figure 2) that differed by the type of information that had to be memorized (object vs. spatial) and by the type of processing necessary for successful performance (maintenance vs. manipulation). The tasks were designed after the experiments that led to the formulation of the above-mentioned theories. Subjects were presented with a series of simple two-dimensional objects presented sequentially at different positions. Maintenance tasks (one-back tasks) were adopted from the classical delayed matching-to-sample task where subjects have to decide whether an object or a position matches the immediately preceding one. Manipulation tasks were designed as two-back tasks, where a stimulus has to be compared to the one presented two items back in the series (D'Esposito, Ballard, et al., 1998; Schumacher et al., 1996; Awh, Smith, & Jonides, 1995; Smith et al., 1995; Cohen et al., 1994). According to D'Esposito, Aguirre, et al. (1998), the two-back tasks require "constant reshuffling of the content held in working memory since different stimuli are simultaneously being stored, inhibited and dropped from memory." Stimulus presentation was lateralized in all tasks

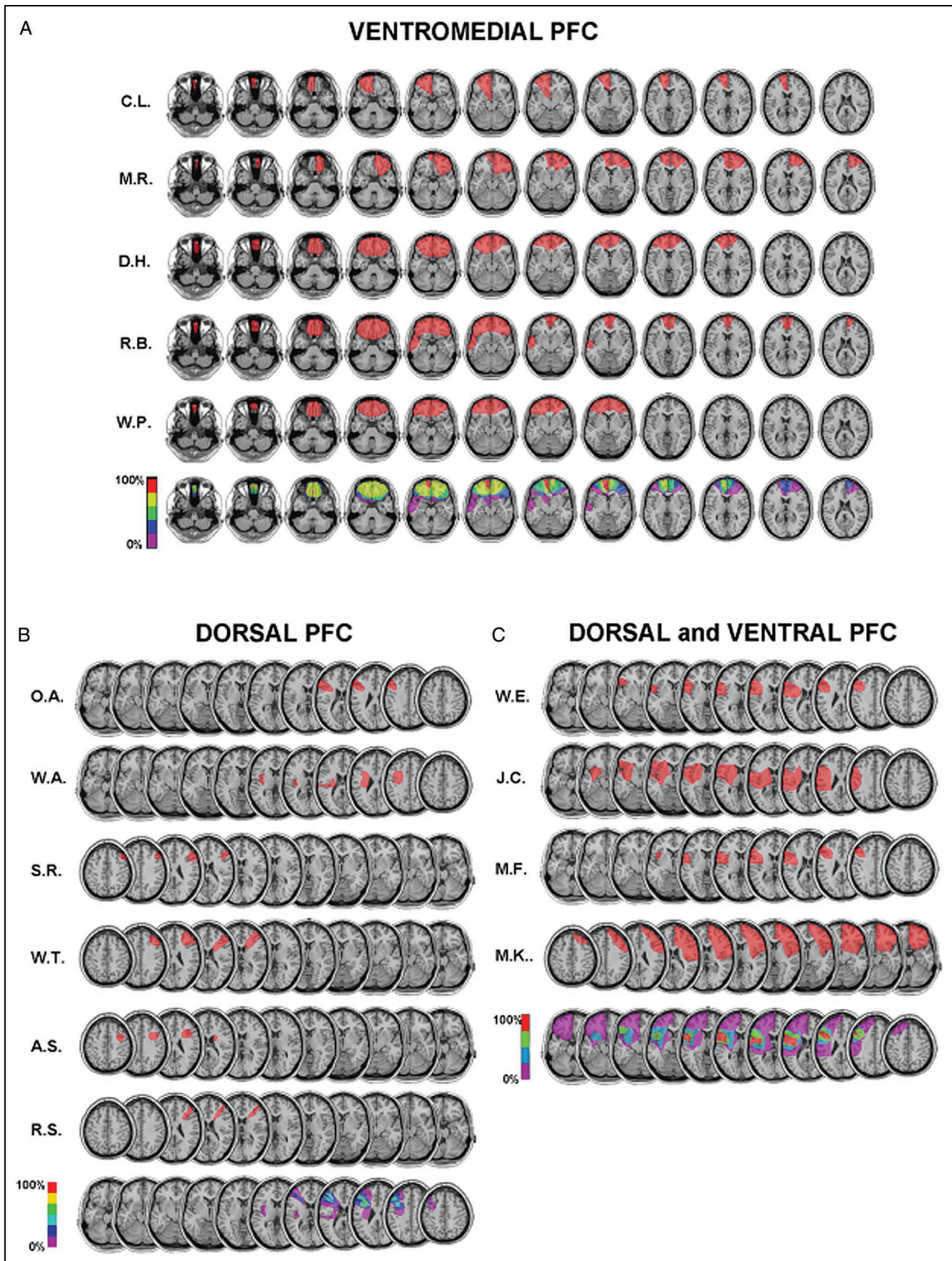


Figure 1. Extent of patients' lesions, reconstructed from CT and MR scans (according to neurological convention, i.e., left is left). The bottom rows present the extent of lesion overlap for each patient group with all unilateral lesions projected on the right hemisphere. A. Lesions of orbitofrontal cortex centered in ventromedial PFC. B. Lesions in the dorsal portion of lateral PFC. C. Lesions involving both dorsal and ventral lateral PFC.

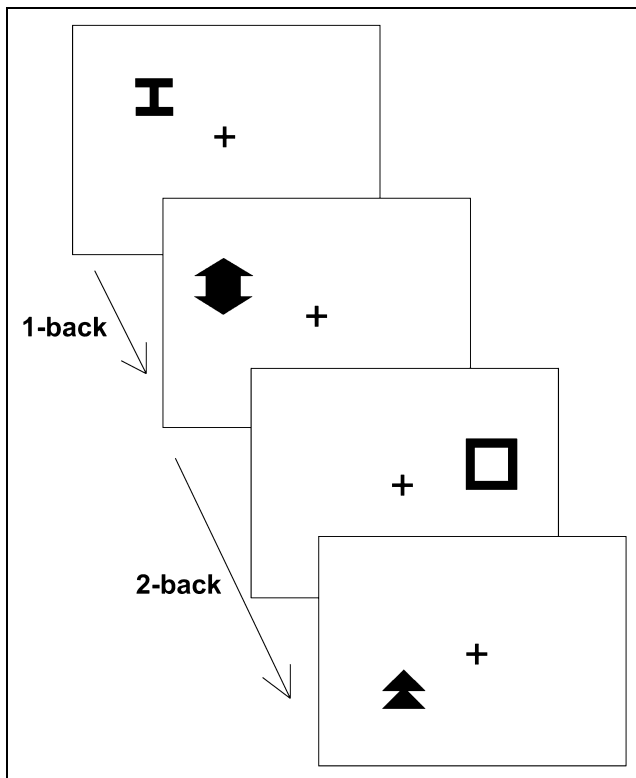


Figure 2. Schematic drawing of the experimental design. Subjects had to decide whether the object's form or position (object vs. spatial) matched the one presented one item or two items back in the series (one- vs. two-back tasks).

to assess hemispheric differences in object and spatial processing (Baker et al., 1996; Owen, Milner, et al., 1996; Smith et al., 1995; Jonides et al., 1993).

RESULTS

Behavioral data from five patients with lesions in the ventromedial PFC (VM group), from six patients with lesions of the dorsolateral PFC (DL group), and from four patients with combined dorsolateral and ventrolateral PFC damage (DVL group) were compared with data from 12 control subjects (see Table 1 for further characteristics of the subjects). The main results are given in Figure 3. ANOVAs were calculated separately for accuracy (in percent)¹ and reaction time (RT) for correct answers (in milliseconds) as dependent measures. Between factors were group (controls, DL, DVL, VM) and within factors with repeated measurements were type of information (object vs. spatial), type of processing (maintenance vs. manipulation), and field of presentation (left vs. right). Whenever a significant effect with the factor "group" occurred, separate ANOVAs for all possible group pairs were performed and additional post hoc *t* tests for each task were calculated between two groups.

Control Subjects

Control subjects made more errors and were slower in the two-back tasks as compared to the one-back tasks [main effect "type of processing": $F(1,11) = 21.79$, $p < .001$ for accuracy and $F(1,11) = 25.33$, $p < .001$ for RT]. Accuracy and RTs did not differ between object and spatial memory tasks [factor "information": $F(1,11) = 1.38$ for accuracy and $F(1,11) = 1.68$ for RT].

No correlation was found between object and spatial one-back tasks (Pearson = 0.44), whereas accuracy in the object and spatial two-back tasks was correlated with each other (Pearson = 0.79, $p < .01$). A correlation was found for the object one- and two-back tasks (Pearson = 0.85, $p < .001$) but not between the spatial one- and two-back tasks (Pearson = 0.29).

Controls versus Patients

For accuracy, we found an overall "group" effect [$F(3,23) = 4.47$, $p < .02$], an interaction between "group" and "type of processing" [$F(3,23) = 6.17$, $p < .004$], and an interaction between "group" and "type of information" [$F(3,23) = 4.03$, $p < .02$]. For RT, there was also a main effect of "group" [$F(2,23) = 4.75$, $p < .02$] and an interaction between "group" and "type of processing" [$F(2,23) = 5.71$, $p < .006$], but no "group" $\times$ "type of information" interaction [$F(2,23) = 0.31$].

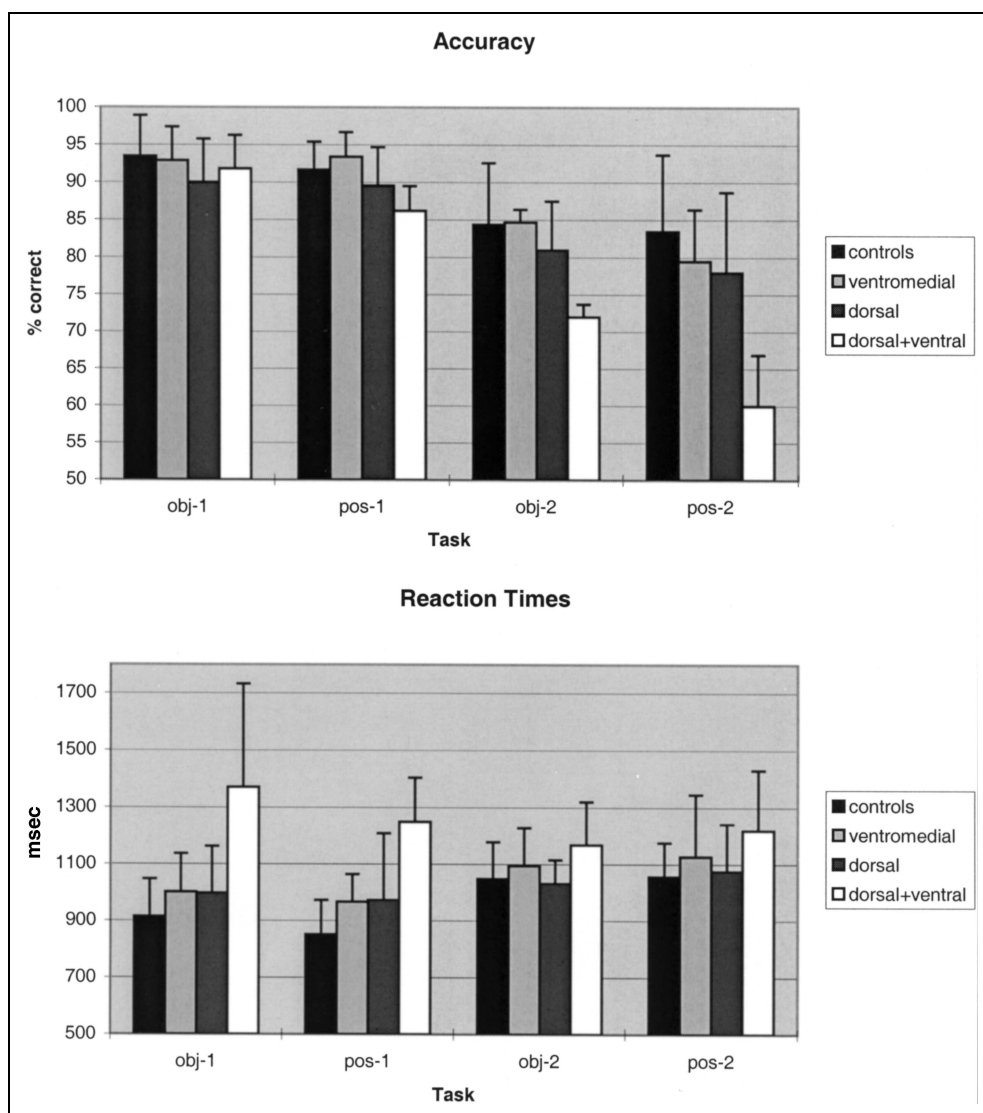
When each patient group was compared to the control group separately, only the DVL group differed significantly. Neither the VM nor the DL group showed reliable differences from the control group ($p > .5$). The DVL versus control comparison for accuracy revealed a main effect of "group" [$F(1,14) = 12.04$, $p < .004$]. As the impairment was more prominent in the two-back tasks, "group" interacted with "type of processing" [$F(1,14) = 17.32$, $p < .001$]. A more pronounced deficit in spatial as compared to object memory led to an interaction between "group" and "type of information" [$F(1,14) = 9.29$, $p < .009$]. Accordingly, post hoc *t* tests revealed significant differences for accuracy in the spatial one-back task ($t = 2.83$, $p < .04$), the spatial two-back task ($t = 5.19$, $p < .001$), and the object two-back task ($t = 4.94$, $p < .02$). Accuracy in the object one-back task did not differ between DVL and control groups ($p > .5$).

For RT, there was a main effect of "group" [$F(1,14) = 14.68$, $p < .002$], due to slower RTs in the DVL group. As RTs were most prolonged in the one-back tasks, there also was an interaction between "group" and "type of processing" [$F(1,14) = 18.13$, $p < .0009$]. Post hoc *t* tests between controls and DVL patients were significant for both one-back tasks ($t = -3.87$, $p < .002$ for the object one-back task, $t = -5.34$, $p < .0001$ for the spatial one-back task). Reaction times in the two-back tasks did not differ between the DVL and control group ($p > .2$).

Table 1. Patient and Control Subject Characteristics

<i>Patient</i>	<i>Sex</i>	<i>Age</i>	<i>Handedness</i>	<i>Education (years)</i>	<i>Lesion Side</i>	<i>Volume (cc)</i>	<i>Etiology</i>	<i>Chronicity (years)</i>
<i>Ventromedial PFC lesions</i>								
C. L.	male	63	ambidextrous	16	left	45	resection of meningioma	10
M. R.	male	44	right	10	bilateral	88	closed head injury	20
D. H.	male	36	right	13	bilateral	79	closed head injury	17
R. B.	male	53	right	12	bilateral	98	closed head injury	23
W. P.	male	54	right	12	bilateral	73	resection of meningioma	0.5
Mean		50.0		12.6		76.6		14.1
<i>Dorsal PFC lesions</i>								
O. A.	male	64	right	13	left	18	stroke	12
W. A.	female	74	right	14	left	26	stroke	10
S. R.	female	77	right	12	right	13	stroke	4
W. T.	male	54	right	18	right	26	resection of cyst	10
A. S.	male	41	right	14	right	14	resection of cyst	6
R. S.	male	35	right	16	right	12	resection of AVM	11
Mean		57.5		14.5		18.2		8.8
<i>Dorsolateral and ventrolateral PFC lesions</i>								
W. E.	male	68	right	14	right	41	stroke	3
J. C.	male	71	right	16	left	103	stroke	9
M. F.	male	65	right	12	left	40	stroke	2
M. K.	male	64	right	18	right	200	aneurysm	17
Mean		67.0		15.0		96.0		7.8
<i>Control subjects</i>								
R. B.	male	57	right	14				
A. A.	male	71	right	12				
J. D.	female	60	right	14				
C. B.	female	68	right	17				
B. H.	male	65	right	17				
G. J.	male	73	right	14				
N. S.	female	78	right	18				
J. B.	female	63	right	12				
J. B.	male	64	right	12				
E. E.	female	79	right	12				
M. A.	male	39	right	14				
W. M.	male	40	right	13				
Mean		63.0		14.1				

Figure 3. Patients performance (percentage of correct response) and RTs for correct responses (in milliseconds) in each of the task employed in the study. Standard deviations are indicated by error bars.



Patient Group Comparisons

The same pattern of results was obtained when the DVL patients were compared either to the VM or the DL patients:

DVL versus DL

Accuracy. Trend for “group” [$F(1,8) = 4.76, p < .06$], interaction “type of processing” $\times$ “group” [$F(1,8) = 11.97, p < .009$], interaction “type of information” $\times$ “group” [$F(1,8) = 11.97, p < .02$], t tests significant for object two-back task ($t = 3.23, p < .018$) and spatial two-back task ($t = 3.22, p < .012$).

RT. Trend for “group” [$F(1,8) = 4.48, p < .07$], interaction “type of processing” $\times$ “group” [$F(1,8) = 5.76, p < .04$], t test trends for object one-back task ($t = -2.16, p < .1$), spatial one-back task ($t = -2.42, p < .06$), due to slower RTs in these tasks.

DVL versus VM

Accuracy. Main effect “group” [$F(1,7) = 22.47, p < .002$], interaction “type of processing” $\times$ “group” [$F(1,7) = 17.37, p < .004$], trend for interaction “type of information” $\times$ “group” [$F(1,7) = 4.97, p < .07$], t tests significant for spatial one-back task ($t = -3.01, p < .02$), object two-back task ($t = -5.29, p < .003$), and spatial two-back task ($t = -4.12, p < .005$).

RT. Interaction “type of processing” $\times$ “group” [$F(1,7) = 7.0, p < .03$], t test for spatial one-back task ($t = 3.18, p < .02$).

Lesion Size

As lesion sizes differed within the groups, nonparametric Mann–Whitney U tests were used for analyses between groups. Lesion size was significantly smaller in the DL group than in the other groups (DL vs. VM: $U = .000$,

$p < .004$, DL vs. DVL: $U = .000$, $p < .01$). There was no difference in lesion size between DVL and VM patients ($U = 10.0$, $p = 1.0$). To further address the effect of lesion size on performance correlation analyses were performed including either all patients, or DVL patients alone (as this was the only group with performance deficits). In both tests no correlation was found between lesion volume and performance (for all patients: Pearson = 0.3, for DVL patients alone: Pearson = 0.4).

Correlation Analyses

To address the question whether the behavioral deficits observed in DVL patients could be related to a common functional deficit, correlation analyses were performed between tasks for this group. We found a correlation for accuracy between the two two-back tasks (Pearson = 0.99, $p < .003$), whereas no reliable correlation occurred between the one-back tasks or between the two spatial or between the two object tasks ($p > .1$).

Effects of Lesion Side

Patients with unilateral lesions were tested for hemispheric specialization for type of stimulus or type of processing. We found no evidence of a specific deficit contralateral to a dorsolateral, dorsolateral and ventrolateral, or ventromedial PFC lesion. Due to group size limitations we also included all patients with unilateral lesions in one ANOVA that comprised the factors "side of lesion" and "presentation side" (ipsi- vs. contralateral to lesion). Lesion side was not distributed equally within the patient groups: the left-side lesions included three DVL patients, two DL patients, and one VM patient. Among the patients with right-sided lesions, only one had DVL lesions, four belonged to the DL group, and none of the VM patients had a lesion restricted to the right side. That is, patients who had shown deficits (group DVL) made a larger contribution to the left side group.

Neither "lesion side" [$F(1,9) = 1.44$] nor "presentation side" (relative to lesion side) had any influence on performance ($F < .1$). Moreover, the "side of lesion" did not interact with "type of processing" [$F(1,9) = 1.63$], nor with "type of information" [$F(1,9) = 1.26$].

DISCUSSION

The aim of this study was to examine the effects of lesions in subregions of the PFC on different components of working memory. We addressed whether PFC can be subdivided anatomically based on the type of information to be held in memory or based on the type of processing necessary for task completion. Models that subdivide PFC based on type of information assert that ventral areas subserve memory for objects and dorsolateral areas subserve spatial memory (Goldman-Rakic,

1995, 1996; Wilson et al., 1993). In contrast, subdivisions based on type of processing claim that both object and spatial information are maintained ventrally, whereas monitoring and manipulation of memory information are carried out more dorsally (D'Esposito, Aguirre, et al., 1998; D'Esposito et al., 1999; Owen, 1997; Owen et al., 1999; Petrides, 1994, 1995, 1996).

Ventromedial PFC Lesions

Patients with lesions in the ventromedial part of the PFC were not impaired in any of the four tasks when compared to controls. These results argue against a role of ventromedial PFC in working memory for object and spatial information.² This is underscored by the fact that four out of five patients had extensive bilateral lesions.

In some imaging studies, activation during working memory tasks was found in an area that overlaps with the lesions in our VM patients. For example, in a memory task for faces, Courtney et al. (1996) found activation at the border of the orbito-frontal and inferior frontal gyri. Goldberg et al. (1996) reported activation of this same region in a spatial working memory task. Also, it has been shown that the orbito-frontal cortex is extensively linked with temporal cortices that are known to process object and face information (Rolls, 1996). Therefore, damage to the orbito-frontal cortex can impair face expression identification (Hornak, Rolls, & Wade, 1996). In our study, however, even those patients with large ventromedial lesions extending to the anterior part of the inferior frontal gyrus and overlapping with the activated regions reported by Courtney et al. and Goldberg et al. did not show impaired object or spatial working memory. Consequently, we assume that the reported activation in the imaging studies did not reflect processes critical to working memory. Instead, ventromedial PFC may be activated in imaging studies of working memory due to its involvement in other processes such as emotional control (O'Doherty, Kringelback, Rolls, Hornak, & Andrews, 2001; Rolls, 1996), decision-making (Bechara, Damasio, Tranel, & Anderson, 1998; Bechara, Damasio, & Damasio, 2000; Bechara, Tranel, & Damasio, 2000; Damasio, Tranel, & Damasio, 1991; Damasio, 1996), and goal-directed behavior (Hollerman, Tremblay, & Schultz, 2000; Eslinger, Grattan, & Geder, 1995). These processes to a certain extent overlap with working memory and are presumably involved in most tasks that intend to test working memory. Our tasks made relatively small demands on these processes (a simple yes/no decision was sufficient), which may explain why lesions of the ventromedial cortex left performance in these tasks intact.

Dorsolateral PFC Lesions

Patients with lesions restricted to dorsal parts of the lateral PFC showed no impairments in the current tasks.

This finding contradicts the predictions of both theories under investigation. The type of information theory predicted impairment in both of the working memory tasks that involved spatial information while the type of processing theory predicted performance deficits in both two-back tasks based on disturbed monitoring capabilities in these patients.

Involvement of dorsolateral PFC in working memory in humans has been mainly derived from imaging studies. In imaging studies, activation of a certain area generally is interpreted to indicate that this area is involved in task completion. However, activation of an area may also reflect control or supervisory processes. This possible confound is usually diminished by subtracting the activity of two tasks that, ideally, differ only in one condition. Nevertheless, activation of an area does not necessarily imply that this area is indispensable for the specific task.

The discrepancy between our results and the imaging data could also be due to the fact that our lesions were unilateral whereas most imaging studies show bilateral activation. Although the latter usually would be taken as evidence that both hemispheres are mandatory for working memory, this not necessarily needs to be the case. Instead, bilateral activation might indicate that the healthy hemisphere is capable of compensating for the loss of its contralateral homologue or that there is redundancy of function whereby each hemisphere can maintain working memory alone. The notion of bilateral PFC capacity is supported by evidence from monkeys with lateral PFC damage (Rossi, Rotter, Desimone, & Ungerleider, 1999). Stimulus detection was only minimally impaired in the visual field contralateral to the PFC lesion. Cutting the corpus callosum increased the contralesional performance deficit suggesting that the PFC-lesioned animal was solving the detection task with the intact hemisphere. However, full recovery of a specific function seems only possible if the lesioned area is not highly specialized for that function. The type of information and type of processing theories under investigation here delineate specific functions to the dorsolateral PFC, yet our patients with selective damage in this area performed normally in tasks designed to assess specifically these functions. Since it seems unlikely that the destruction of this area can leave functions that the dorsolateral PFC is specialized for completely intact, we propose a less precise role of dorsolateral PFC in working memory than that suggested by recent models. This assumption is supported by imaging studies that found dorsolateral PFC activation in tasks that neither involved spatial information (Cohen et al., 1997; Courtney et al., 1996; Smith et al., 1996) nor manipulation of information in working memory (Rypma, Prabhakaran, Desmond, Glover, & Gabrieli, 1999; Manoach et al., 1997). In 1942, Malmö reported that memory in monkeys with PFC lesions

remained intact—as long as the animals were not distracted during the delay. This further challenges a specific role of PFC in spatial working memory (see below for further considerations).

In contrast to our results, Baldo and Shimamura (2000) and Ptito, Crane, Leonard, Amsel, and Caramanos (1995) reported spatial memory deficits in patients whose lesions invaded BA 46, which constitutes a major component of the dorsolateral PFC. However, inspection of their lesion data suggests that a considerable part of their patients had lesions that extended ventrally and posteriorly. Thus, the lesion patterns resemble our DVL group. In the DVL group, where the lesions within lateral PFC extended from the ventral to the dorsolateral region, spatial memory was indeed impaired. The fact that most studies did not control systematically for involvement of areas adjacent to the dorsolateral PFC may explain the diversity of results reported after prefrontal lesions in humans.

Combined Dorsal and Ventral Lesions in the Lateral PFC

In contrast to the other patient groups, patients with lesions involving both the ventral and dorsal parts of lateral PFC revealed impaired performance in all tasks employed in the current study. Although lesions in the DVL group were larger as in the DL group, this deficit cannot simply be attributed to lesion size, as even those DVL patients with smaller lesions (W. E. and M. F.) showed severe impairment whereas patients of the other groups with large lesions were not impaired. Further, no correlation between lesion size and performance occurred. Thus, the involvement of both dorsal and ventral lateral PFC rather than lesion size per se seems to be crucial for the observed deficits. Another explanation would hold that ventrolateral PFC alone maintains working memory. This assumption would presume that all working memory functions—irrespective of type of information and type of processing—are located within this single brain area. This clearly contradicts the findings of all previous studies on the neural correlates of working memory. However, since we were unable to test patients with isolated lesions in ventrolateral PFC, we cannot rule out this possibility completely. Interestingly, the intact PFC was not able to compensate for extensive unilateral lesions. This suggests that the intact performance in the restricted DL lesion group might have been due in part to intrahemispheric compensation from the intact ventral PFC in the lesioned hemisphere. This would argue for a more distributed functional capacity of lateral PFC.

The DVL patients were slower in both one-back tasks, slightly less accurate in the spatial one-back task, and showed severe accuracy deficits in both two-back tasks. The differences in RT versus accuracy impairment between the one- and two-back tasks could reflect

changes in strategy, where accuracy was stressed in the one-back tasks and speed was stressed in the two-back tasks. However, changes of strategy are unlikely to evolve in such a systematic pattern unless caused by a common deficit. We suggest that the fast responses in the two-back tasks were due to the patients realizing that they could not respond accurately, and thus instead focused on fast responses. In contrast, during the one-back tasks they were able to partially compensate for their deficits by slowing down their responses. Thus, the changes in strategy may have been secondary to memory deficits.

With respect to information type, the results support some lateral PFC specialization for spatial information processing, in the absence of subregional specialization within lateral PFC. Since control subjects did not show performance differences between object and spatial tasks, the information-specific deficits in the DVL patients cannot be easily attributed to variances in task difficulty.

The type of processing theory predicted deficits in patients with extensive lateral prefrontal damage for both the one- and the two-back tasks because of combined damage to the ventral “maintaining center” and the dorsal “monitoring center.” In our study, these patients showed severe deficits in both two-back tasks supporting a crucial role of lateral PFC for monitoring of information in working memory. Also, we found a high correlation between patients’ performance in the object and spatial two-back tasks suggesting that a common neural substrate within lateral PFC may underlie these deficits. In accordance with the type of processing theory, these results argue for a common network within lateral PFC that subserves monitoring for both object and spatial information within working memory.

The type of processing theory also implies a hierarchical organization of processes and their anatomical substrates in PFC where monitoring in dorsal areas requires prior maintaining in ventral areas (Owen, 1997; Owen et al., 1999). Two-back tasks require both maintaining and monitoring of information. Thus, lesions that involve the proposed neuronal correlates of both processes should especially affect performance in the two-back tasks—as was the case in our DVL patients. Performance was not nearly as severely affected in the one-back tasks. Nevertheless, the prolonged RTs in the two one-back tasks could be interpreted in terms of dysfunction in the common “maintaining” center. However, the concept of a common maintaining center cannot explain why spatial and object processing were differently affected in our patients and why performance in the 2 one-back tasks did not correlate.

General Discussion

Taken together, the performance of the three groups of patients with damage in subregions of PFC supports a

critical role of lateral PFC in working memory. However, their performance was inconsistent with a number of the predictions of both the type of information and the type of processing theory.

First, we found that ventromedial and—more surprisingly—dorsolateral PFC lesions left working memory intact. This finding is not in accord with models that ascribe a specific function inherent in our tasks to one of these brain areas. The idea that lateral PFC might be less specifically organized is supported by fMRI studies that found activation in dorsolateral area during tasks that neither required memory for spatial information nor monitoring of memory content. For example, Rypma et al. (1999) found activation in the ventrolateral PFC when subjects had to maintain one or three letters in working memory and additional activity in the dorsolateral PFC when retention of six letters was required—although no overt manipulation of information was necessary. A similar finding was reported by Manoach et al. (1997) who compared memory for digits—the dorsolateral PFC showed activation when five but not when two digits had to be remembered. Baker et al. (1996) reported dorsolateral PFC activation in both object and spatial memory tasks when they involved a long retention delay. Others observed prefrontal activation when two competing tasks had to be worked on simultaneously, but not when one task was performed alone (D’Esposito et al., 1995; Corbetta, Miezin, Dobmeyer, Shulman, & Petersen, 1991), or when self-ordering of information was requested (Smith et al., 1996). D’Esposito, Ballard, Aguirre, & Zarahn (1998) found lateral PFC activation in nonworking memory tasks, only to a lesser extent as compared to working memory tasks. These findings suggest that dorsolateral PFC activation might have been observed in a number of tasks that did not require manipulation or other high-order cognitive functions if the subtraction method typically employed in functional imaging had not concealed this activity. Taken together, the common feature of the tasks that led to dorsolateral PFC activation seems to be that they were cognitively demanding—regardless of whether they required monitoring or spatial processing. Overall, the functional imaging data suggest that the dorsolateral area is part of a larger network in the lateral PFC, which is progressively recruited with increasing task demands. The current data further suggest that lesions that leave considerable parts of this lateral PFC network intact can be compensated for behaviorally.

Another finding from our study that is not in accord with the predictions of current theories of PFC function is that all patients, even those with extensive lesions in lateral PFC, were performing well above chance level in both maintenance tasks. Thus, a considerable part of maintaining information in memory seems to depend on either non-PFC structures or homologous regions of one intact hemisphere (Postle & D’Esposito, 1999).

Indeed, preserved recognition memory for different types of stimuli after frontal lobe lesions has been reported earlier and was contrasted to deficits in free recall, associative memory, source memory, and memory for temporal order (Dimitrov et al., 1999; Kopelman, Stanhope, & Kingsley, 1997; Rushworth, Nixon, Eacott, & Passingham, 1997; Owen et al., 1995; Shimamura, Janowsky, & Squire, 1990; Janowsky, Shimamura, Kritchevsky, & Squire, 1989; Milner et al., 1985). On the other hand, both human and nonhuman primate literatures report memory deficits for objects after temporal lobe lesions (Owen, Morris, et al., 1996; Owen et al., 1995; Mishkin & Ungerleider, 1982) and suggest a crucial role of the superior parietal lobe in spatial memory (Ungerleider & Mishkin, 1982). Petrides (1995, 1996) argued that posterior brain regions subserve automatic recognition based on familiarity, whereas PFC would only be involved in active retrieval of information from memory, such as in free recall. With respect to our study, the good performance of all patients in the one-back tasks could be explained by intact “automatic” recognition in posterior cortices.

Lateral PFC may become more crucial in supporting maintenance in more demanding circumstances where posterior regions might no longer be sufficient to support this process alone. In accord with this assumption, in recent fMRI studies, a linear relationship between activation of frontal areas and memory load has been described (Rypma et al., 1999; Braver et al., 1997; Cohen et al., 1997; Manoach et al., 1997). In order to support maintenance in PFC, information processed posteriorly has to reach the frontal cortex and corresponding pathways from temporal objects and parietal space areas have to be identified (Friedman & Goldman-Rakic, 1994; Quintana & Fuster, 1993; Wilson et al., 1993; Barbas, 1988; Batuev et al., 1985; Kawamura & Naito, 1984). Information-dependent delay activity observed in PFC observed during single-unit recording studies (Wilson et al., 1993; Funahashi et al., 1989) may be explained against this background.

The different contributions of posterior and frontal brain structures to maintaining information in memory are further illustrated by studies of Miller, Erickson, and Desimone (1996) who found context-dependent differences in neural activity in temporal and frontal cortices. They recorded parallel delay activity in temporal and frontal cortices in a recognition task in nonhuman primates, which supports the involvement of both brain areas in maintaining information in memory. However, responses in PFC were less often stimulus-selective but conveyed more information about whether a given test stimulus was a match to the sample. Furthermore, sample-selective delay activity in PFC cortex was—unlike in temporal cortex—maintained throughout the trial even when other test stimuli intervened during the delay. Another specific function of PFC seems to involve the integration of object and spatial information (Prab-

hakaran, Narayanan, Zhao, & Gabrieli, 2000; Sinha, Brodsky, Willing, Pellouchoud, & Knight, 2000). Some neurons in PFC produce delay activity for both objects and locations and, more importantly, for certain combinations of objects and locations (Rao et al., 1997).

Our results do suggest a crucial role of lateral PFC in monitoring and retrieval. However, these terms—introduced by Petrides (1995, 1996)—summarize a number of subprocesses making it difficult to discern which subprocess activates PFC. Nevertheless, “manipulation” can be operationally defined as a process that is not involved in maintenance tasks and selectively leads to activation of the dorsolateral PFC. This dichotomy may not be as sharp as proposed (see also Smith & Jonides, 1999), as even one-back tasks require a number of demanding processing steps: For each stimulus, subjects must determine the relevant information, compare that information with the information encoded for the immediately preceding stimulus while ignoring information encoded prior to one-back, commit the current information to memory for use with the subsequent stimulus, and, at the same time, ignore the irrelevant information (spatial information in the object task, object information in the spatial task), and remember the button assignment for responding. On the other hand, a high memory load, such as in the Rypma et al. (1999) study, not only increases the number of items to be stored in memory, but also requires better monitoring and organization of memory. Thus, we propose that there is no strict dichotomy between maintenance/recognition and manipulation/monitoring. Rather, maintenance and manipulation/monitoring are inextricably linked.

Conclusions

Instead of discrete and distinctive memory processes, we propose a continuous spectrum of memory support functions where maintaining and monitoring demands are intermixed. Likewise, our results also argue against a strict compartmentalization of the anatomical substrates that subserve these processes within lateral PFC. Rather, ventral and dorsal areas of the lateral PFC network may be recruited successively with increasing task demands and compensation for small unilateral lesions is possible as is clearly evident.

METHODS

Subjects

Fifteen patients with lesions in the PFC and 12 neurologically normal controls (mean age = 60 + 18/–25 years) participated in this study. Patients were assigned to one of three groups. Patient group VM included five patients with lesions in the ventromedial PFC (BA 10, 11, 12). Out of these, two had additional lesions in the

ventral-most part of the inferior frontal gyrus (BA 47). Four had extensive bilateral lesions, one had a left side lesion only. The DL group comprised six patients with lesions only in the dorsolateral PFC (BA 9, 46, 9/46), whereas in the DVL group of four patients the lesions encompassed both the dorsolateral and the ventrolateral PFC (BA 9, 46, 9/46, 45, 47).

Patients were selected from the outpatient neurological clinic of the Veterans Medical Administration Center in Martinez, CA. After identifying their lesions from CT or MRI scans obtained in the chronic state of disease, patients were assigned to one of the three groups by author R. T. K. For all patients, the lesions occurred between 6 months and 23 years prior to testing. No patient had a history of other significant medical diseases such as psychiatric disorder, dementia, substance abuse, or additional neurological events. For further information on the patients, refer to Table 1.

All lesions were transcribed from the original scans onto corresponding axial templates (Frey, Woods, Knight, Scabini, & Clayworth, 1987). Software permitted group averaging, determination of lesion volume, and plotting of the reconstructed data.

Design

In all tasks, subjects fixated on a central cross while a series of abstract blue colored 2-D objects appeared against a light gray background on a virtual circle, 6° from fixation. Each object was chosen from a set of 32 objects and positioned at 1 of 14 equally spaced locations. Four different tasks were employed:

In the **one-back object task**, subjects had to decide whether the form of the presented object matched the immediately preceding one. In the **one-back spatial task**, subjects had to decide whether the position of the presented object was identical to the position of the preceding object. In both one-back tasks, each object appeared for 100 msec followed by a 4250-msec delay. In the **two-back object task**, the object's form had to be compared to the one presented two items back in the sequence whereas in the **two-back spatial task** the location of the object on the screen had to be compared to the one two items back in the series. Each object appeared for 250 msec followed by a 2000-msec delay, for both two-back tasks.³ Thus, the delay interval between the to-be-compared stimuli was equivalent for the one- and the two-back tasks (4250 msec).

Each experimental session was comprised of four blocks, one for each task, with a 5-min break between blocks. A block consisted of 12 sequences of either 10 (one-back tasks) or 20 stimuli (two-back tasks) presented serially. A mouse button press was required on every stimulus (apart from the first in the one-back tasks and the first and second in the two-back tasks), with the right button meaning "same" and the left meaning "differ-

ent." In object tasks, a repeated object could either appear at the same position as the stimulus it had to be compared to or in a new position, both cases requiring a "same" decision. Likewise, in the spatial tasks, a repeated position ("same") could either be represented by a repeated or a new object. However, repeated objects and positions never shifted between the visual hemifields in order to avoid interference with lesion lateralization. Reaction time was limited to 2 sec and subjects were asked to respond "as quickly and as accurately" as they could. All subjects completed the maintenance tasks before the manipulation tasks, but the order of the object and spatial versions was counterbalanced across subjects. Sequences in the object and spatial tasks were randomized. Subjects sat in a comfortable chair in a dimly lit and sound-attenuated room. Their viewing distance from the 50.8-cm computer monitor was 1.2 m. Objects on the screen were approximately 5 × 5 cm large. The virtual circle on which the objects were presented had a diameter of 25 cm with equal distances between positions.

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Notes

1. Analyses with corrected hit rates, where the false alarms were subtracted from the hits, confirmed the following results and are not further reported.
2. It should be noted that this does not eliminate the possibility that ventromedial PFC is involved in working memory for emotional stimuli.
3. The presentation rate was increased relative to the one-back tasks, as pilot experiments had shown that even controls were hardly capable of solving the two-back tasks with shorter presentation times.

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